Part B

1 OCN

Total Delay = Propagation + Transmition + queing + processing

~~Propagation~~ Transmition = $\frac{\text{Packet size (bits)}}{\text{Link bandwidth (b/s)}}$

$= \frac{100 \text{ bytes} \times 8}{4 \times 10^{9} \text{ bps}}$

$= 2 \times 10^{-7}$ s

~~Transmission~~ Propagation = $\frac{\text{distance (m)}}{\text{Signal Speed (m/s)}}$

$= \frac{0.5}{100} \text{ m} \times \frac{1}{200{,}000 \times 1000 \text{ m/s}}$

$= 2.5 \times 10^{-11}$ s

queing = 0s

processing = $0 + 0.5 \times 100$

$= 50$ ns

$= 50 \times 10^{-9}$ s

Total delay = 2.5×10^{-11} s $+ 2 \times 10^{-7}$ s $+ 50 \times 10^{-9}$ s

$= 2.5 \times 10^{-7}$ s

% Propagation = $\frac{2.5 \times 10^{-11}}{2.5 \times 10^{-7}} \times 100$

$= 0.01$ %

SAN

total delay = propagation + Transmission + queuing + processing

$$\text{Propagation} = \frac{\text{Distance (m)}}{\text{Signal Speed (m/s)}}$$

$$= \frac{5\ m}{2 \times 10^{8}\ m/s}$$

$$= 2.5 \times 10^{-8}\ s$$

$$\text{Transmission} = \frac{\text{Packet size (bits)}}{\text{link data rate bps}}$$

$$= \frac{100 \times 8\ b}{4 \times 10^{8}\ b/s}$$

$$= 2 \times 10^{-7}\ s$$

queuing = 0

$$\text{processing} = x + (0.5\ ns/\text{byte})$$

$$= 0.3 \times 10^{-6}\ s + 0.5 \times 100\ ns$$

$$= 0.3 \times 10^{-6}\ s + 50 \times 10^{-9}\ s$$

$$= 3.5 \times 10^{-7}\ s$$

$$\text{Total delay} = 2.5 \times 10^{-8} + 2 \times 10^{-7} + 3.5 \times 10^{-7}$$

$$= 5.75 \times 10^{-7}\ s$$

$$\%\ \text{propagation} = \frac{2.5 \times 10^{-8}\ s}{5.75 \times 10^{-7}\ s} \times 100$$

$$= 4.34\ \%$$

LAN

Total delay = propagation + Transmission + queuing + processing

$$\text{Propagation} = \frac{\text{Distance (m)}}{\text{Signal Speed (m/s)}}$$

$$= \frac{5000\ m}{2 \times 10^{6}\ ms^{-1}}$$

$$= 2.5 \times 10^{-5}\ s$$

$$\text{Transmission} = \frac{\text{Packet size (bits)}}{\text{link data rate b/s}}$$

$$= \frac{100 \times 8\ b}{4 \times 10^{9}\ b/s}$$

$$= 2 \times 10^{-7}\ s$$

queuing = 0

$$\text{processing} = x + (0.5\,ns / \text{byte})$$

$$= 3 \times 10^{-6}\ s + 0.5 \times 100\ ns$$

$$= 3 \times 10^{-6}\ s + 50 \times 10^{-9}\ s$$

$$= 3.05 \times 10^{-6}\ s$$

$$\text{Total delay} = 2.5 \times 10^{-5} + 2 \times 10^{-7} + 3.05 \times 10^{-6}$$

$$= 2.825 \times 10^{-5}\ s$$

$$\%\ \text{propagation} = \frac{2.5 \times 10^{-5}}{2.825 \times 10^{-5}} \times 100$$

$$= 88.5\ \%$$

WAN

Total delay = Propagation + Transmission + queing + processing

$$\text{Propagation} = \frac{\text{Distance (m)}}{\text{Signal speed (m/s)}}$$

$$= \frac{5000 \times 1000 \text{ m}}{2 \times 10^{8} \text{ ms}^{-1}}$$

$$= 2.5 \times 10^{-2}$$

$$\text{Transmission} = \frac{\text{data packet (bits)}}{\text{Data link rate b/s}}$$

$$= \frac{100 \times 8 \text{ b}}{4 \times 10^{9} \text{ b/s}}$$

$$= 2.5 \times 10^{-7} \text{ s}$$

queing = 0

$$\text{processing} = x + (0.5 \text{ ns/byte})$$

$$= 30 \times 10^{-6} + (0.5 \times 100 \text{ ns})$$

$$= 30 \times 10^{-6} + 50 \times 10^{-9}$$

$$= 3.005 \times 10^{-5}$$

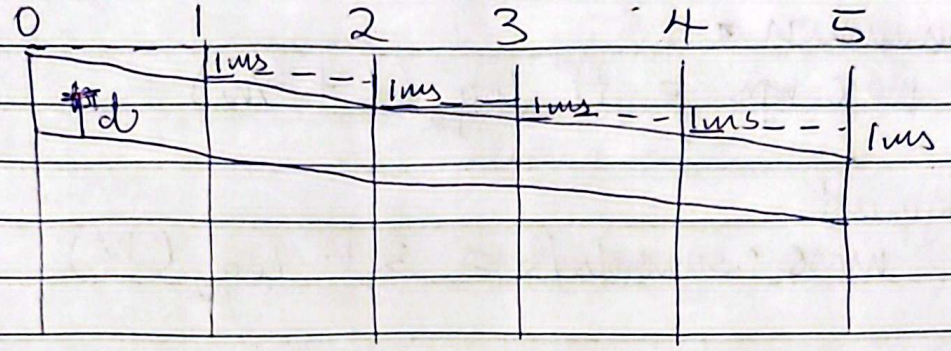
$$\text{Total delay} = 2.5 \times 10^{-2} + 2.5 \times 10^{-7} + 3.005 \times 10^{-5}$$

$$= 2.50303 \times 10^{-2}$$

$$\% \text{ propagation} = \frac{2.5 \times 10^{-2}}{2.50303 \times 10^{-2}} \times 100$$

$$= 99.8\%$$

2



Total delay = 5 (propagation + Transmission)
+ 0 + 0

propagation = 1ms

Transmission = $\frac{\text{Data packet (bits)}}{\text{Data link rate b/s}}$

$= \frac{1000 \times 8 \text{ bits}}{100 \times 10^6 \text{ b/s}}$

$= 8 \times 10^{-5}$ s

Total delay = $5(1 \times 10^{-6} + 8 \times 10^{-5})$

$= \cancel{4.05 \times 10^{-4}}$ s $5(8.1 \times 10^{-5})$

$= 4.05 \times 10^{-4}$ s

For $n = 1$ $= 1 \times 4.05 \times 10^{-4}$ s

$= 4.05 \times 10^{-4}$ s

$n = 12$ $= 12 \times 4.05 \times 10^{-4}$

$= 4.86 \times 10^{-3}$ s

3

Shannon

max bps $= H \log_2(1 + S/N)$

Nyquist

max symbols/s $= 2H \log_2(M)$

$H = 1\,MHz$
$= 10^6\,Hz$

$S/N = 7$

M (signal levels) $= 2$

a Each ~~bracket~~ Symbol 2 possible values

Shannon $- 10^6 \log_2(1+7)$
$- 10^6 \log_2 8$
$= 3 \times 10^6$ bps

Nyquist $= 2 \times 10^6 \log_2(2)$

$= 2 \times 10^6$ symbols / second

Nyquist has tighter upper bound

b 16 possible values

Shannon $= 10^6 \log_2(1+7)$
$= 3 \times 10^6$ bps

Nyquest $= 2 \times 10^6 \log_2(16$
$= 8 \times 10^6$

Shannon has a tighter upper bound

4

a Rate per session $= \dfrac{\text{Total link rate}}{\text{Number of sessions}}$

$$= \frac{100 \times 10^6 \text{ bps}}{100}$$

$$= 10^6 \text{ bps}$$

b

~~Pa~~ achievable rate per session $= \dfrac{\cancel{100}\ \text{Session}}{1 + (N-1)\dfrac{B}{B+1}}$

$$= \frac{100}{1 + 99 \cdot \dfrac{B}{B+1}}$$

$B = 10s$

$$R_{avg} = \frac{100}{1 + 99 \times \frac{10}{11}} = \frac{100}{90.1} = 1.11 \text{ Mbs}$$

$B = 1s$

$$R_{avg} = \frac{100}{1 + 99 \times \frac{1}{2}} = \frac{100}{50.5} = 1.98 \text{ Mbs}$$

$B = 0.1s$

$$R_{avg} = \frac{100}{1 + 99 \times \frac{0.1}{1.1}} = \frac{100}{10} = 10 \text{ Mbs}$$

$B = 0.01$

$$R_{avg} = \frac{100}{1 + 99 \times \frac{0.01}{1.01}} = \frac{100}{1.98} = 50.5 \text{ Mbs}$$